

# Troubleshooting

# 13

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The cat cracker must operate reliably and efficiently. It must also operate safely and comply with federal, state, and local environmental requirements. A typical FCC unit circulates tons of catalyst per minute, processes various types of feedstock, and uses hundreds of control loops, any of which could make operation difficult. Proper troubleshooting will ensure that the unit operates at maximum reliability and efficiency, while complying with environmental concerns.

Troubleshooting deals with identifying and solving problems. Problems can be immediate or long term. They can be off-spec products, poor efficiency, equipment malfunction or environmental excursion. Problems can be related to startup issues, instrumentation, loss of utilities, equipment wear, changes in the operating conditions, and operator errors.

This chapter outlines fundamental steps toward effective troubleshooting. It provides a practical and systematic approach to develop a solution. General guidelines are provided for identifying problems and determining a diagnosis. It is written with the unit process engineer in mind. No matter where the problem originates, he/she will be the point person for solving it.

Before beginning to troubleshoot, one must understand the unit's "normal" operating mode and be able to list several leading indicators to confirm the operating baseline of the unit. For example, what supporting evidences are there that the operation of the FCC unit is:

- Safe
- Clean (environmentally and meeting product specifications)
- Stable
- Operating within its maximum or minimum limits

Once an abnormal situation occurs, effective troubleshooting starts by addressing the following questions:

- What is the "leading indicator" of an issue?
- What are some of the evidences that confirm this "abnormal issue?"
- What resources (for example DCS data/trends, lab data analyses, outside operator's observance, etc) are available to troubleshoot the abnormal situation?
- What could be causing the problem (rank by importance, with 1 being most important)?
- Where or what one would look at to diagnose the problem?
- What corrective actions one would take to resolve this issue?
- Are there proactive actions that can be taken to prevent occurrences of this problem in the future?
- What would have been possible consequences of no or delayed response to the symptom/problem?

Long-term solutions can include improved operating procedures, scheduled training, preventative maintenance, and installation of new equipment or controls.

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## 13.1 Several general guidelines for effective troubleshooting

A successful troubleshooting assignment will require someone to:

- Be a good listener.
- Know the "normal" operating parameters.
- Gather historical background.
- Evaluate "common" and "uncommon" causes of problems.
- Examine goals and constraints to verify the applicability of the present operation.

Management, engineering, and operations departments perceive problems differently. Frequently, there is someone familiar with the operation that most likely knows the symptoms, and possibly can offer a solution to the problem, but for various reasons, people who are in a position to implement the solution may not have thought to ask that person. Typically, those closest to the problem are the unit operators and maintenance foremen, and they will offer the most valuable input. All four operating shifts need to be consulted. Do not draw any conclusions before gathering all applicable facts.

Examine similar problems that have occurred previously in the system to determine how they were diagnosed and solved. Review the operating and maintenance records. Compare the performance of the normally operating unit to the current problematic operation. Make sure that all the unit trends are current, including catalyst data, and the heat and weight balance data. Note any changes that might relate to the problem. Reliable historical data always helps to identify and diagnose problems.

Begin by listing all potential causes or combinations of causes, using a “brainstorming” approach. Then, systematically rule out each cause. Do not eliminate uncommon causes too quickly; if it were an easy problem someone would have already taken care of it. Additionally, ensure that the limits outlined by process and equipment documentation are consistent with the actual operation of the unit.

Most FCC problems are due to changes in the feedstock, catalyst, operating variables, and/or mechanical equipment. As stated previously, the solution can take the form of improving yields, avoiding shutdowns or, increasing unit reliability.

The majority of my troubleshooting assignments have been related to catalyst circulation issues, excessive catalyst loss, above average afterburning, premature coking, high CO/NO<sub>x</sub> emission and “abnormal” products quality/quantity.

Regarding catalyst circulation issues, it has been my experience that understanding the key physical properties of the FCC catalyst and unit’s pressure will go long way in solving limitations and/or erratic catalyst circulation. Consequently, the next two (2) sections contain fundamentals of physical properties, pressure balance, and catalyst circulation.

---

## 13.2 Key aspects of FCC catalyst physical properties

- FCC catalyst is comprised of “micro-sphere” particles.
- 97% of the particle size distribution ranges from 0.5  $\mu\text{m}$  to 150  $\mu\text{m}$ .
- The FCC catalyst’s density is reported to be relative to water density.
- Water density is 62.4 pound per cubic feet (@ 60 °F) or one gram per cubic centimeter (g/cc).
- The density of the “as shipped” fresh FCC catalyst is always less than the water density.
- As shipped, the fresh catalyst’s density typically ranges from 65% to 85% of the water density (0.65 g/cc to 0.85 g/cc).
- The density of E-cat (spent) catalyst is also virtually less than water. In some instances, it can be slightly higher.
- A dense phase fluidized catalyst bed looks very much like a boiling liquid and shows “liquid-like” behavior. Fluidized solids will flow like a liquid from vessel to vessel. This is the basic concept for the operation of the FCC Unit.
- For a catalyst to flow like water, the forces must be transmitted through catalyst particles and not to the vessel wall.
- Air, fuel gas, nitrogen, and steam are commonly used to help in the fluidization, or aeration, of the catalyst. However, they must be dry.
- The lowest superficial gas velocity, in which the pressure drop across a fixed-bed of catalyst equals the weight of the bed, is referred as incipient fluidization velocity or minimum fluidization velocity. Any slight increase in the gas velocity will cause incremental lifting or expansion of the catalyst bed. The velocity in which gas bubbles are first observed is known as “minimum bubbling velocity.”
- The presence of fines in the particle size distribution is helpful for fluidization. The fines act as a lubricant for the larger particles. These smaller particles move more easily in the gas.
- De-aeration is the loss of fluidity to a packed bed. The fines content, as well as the shape of the catalyst, affects the de-aeration rate.
- The ratio of minimum bubbling velocity to minimum fluidization velocity provides a useful tool to assess the fluidity of the FCC catalyst.
- The catalyst’s particle size distribution, its shape, and particle density play key roles in its ability to be fluidized.

### 13.3 Fundamentals of catalyst circulation

An FCC unit is a “pressure balance” operation, basically behaving similar to a water manometer. Differential pressure between the regenerator and reactor vessels is the driving force that allows for the fluidized catalyst to circulate between the regenerator and reactor vessels (see Fig. 13.1 for a typical pressure balance). The slide or butterfly valve located in the regenerator flue gas line is used to regulate the differential pressure between the regenerator and reactor vessels. The reactor pressure is often controlled by the wet gas compressor though there FCC units in which the Main Fractionator Tower top is controlled independently.

Fresh catalyst is added to make-up for the catalyst losses from the reactor/regenerator vessels, as well as to compensate for the loss of catalyst activity. The catalyst inventory in the unit is controlled by periodic withdrawal of the excess catalyst from the regenerator vessel.

The catalyst level in the catalyst stripper vessel is controlled by a slide or plug valve located in the spent catalyst standpipe. In most FCC units, the cracking temperature is controlled by regulating the catalyst flow from the regenerator via slide or plug valve that are located in the regenerated catalyst standpipe. In Model IV and Flexicracker FCC units, the differential pressure between the reactor and regenerator is the primary control point for regulating catalyst circulation from the regenerator to the reactor.

In FCC regenerators that operate in **partial combustion mode** of catalyst regeneration, the combustion air rate is regulated to target a given concentration of carbon monoxide (CO) in the regenerator flue gas and/or a set level of coke on the regenerated catalyst (CRC).

In FCC regenerators that operate in **full burn mode** of catalyst regeneration, an excess concentration of oxygen is maintained in the regenerator flue gas to ensure complete combustion of carbon monoxide (CO) to carbon dioxide (CO<sub>2</sub>).

Catalyst “raw” level in the regenerator is determined by measuring the differential pressure between the pressure above the air distributor and the regenerator dilute/top pressure. There is often another pressure tap, about 5 feet (158 cm) above the air distributor that is used to measure catalyst flowing density.

In the reactor/stripper, the “raw” catalyst level is determined by measuring the differential pressure from the catalyst stripper bottom versus the reactor top pressure. The actual catalyst level can be calculated by employing the catalyst density readings in the catalyst stripper.

Catalyst circulation rate is dependent on the following parameters:

- Fresh feed rate.
- Use of naphtha, LCO, HCO, or slurry recycle to the riser.
- Cracking temperature.
- Feed temperature to the riser.
- Reactor and regenerator pressures.
- Regenerator dense bed temperature.

The regenerator dense bed temperature is dependent on:

- Feed quality.
- Fresh catalyst addition rate and/or its activity.
- Ambient condition and air blower discharge temperature.

- Catalyst cooler duty and/or other removal schemes.
- Performance efficiencies of feed nozzles and catalyst stripping.
- Level of afterburning.
- Concentration of CO in the regenerator flue gas.

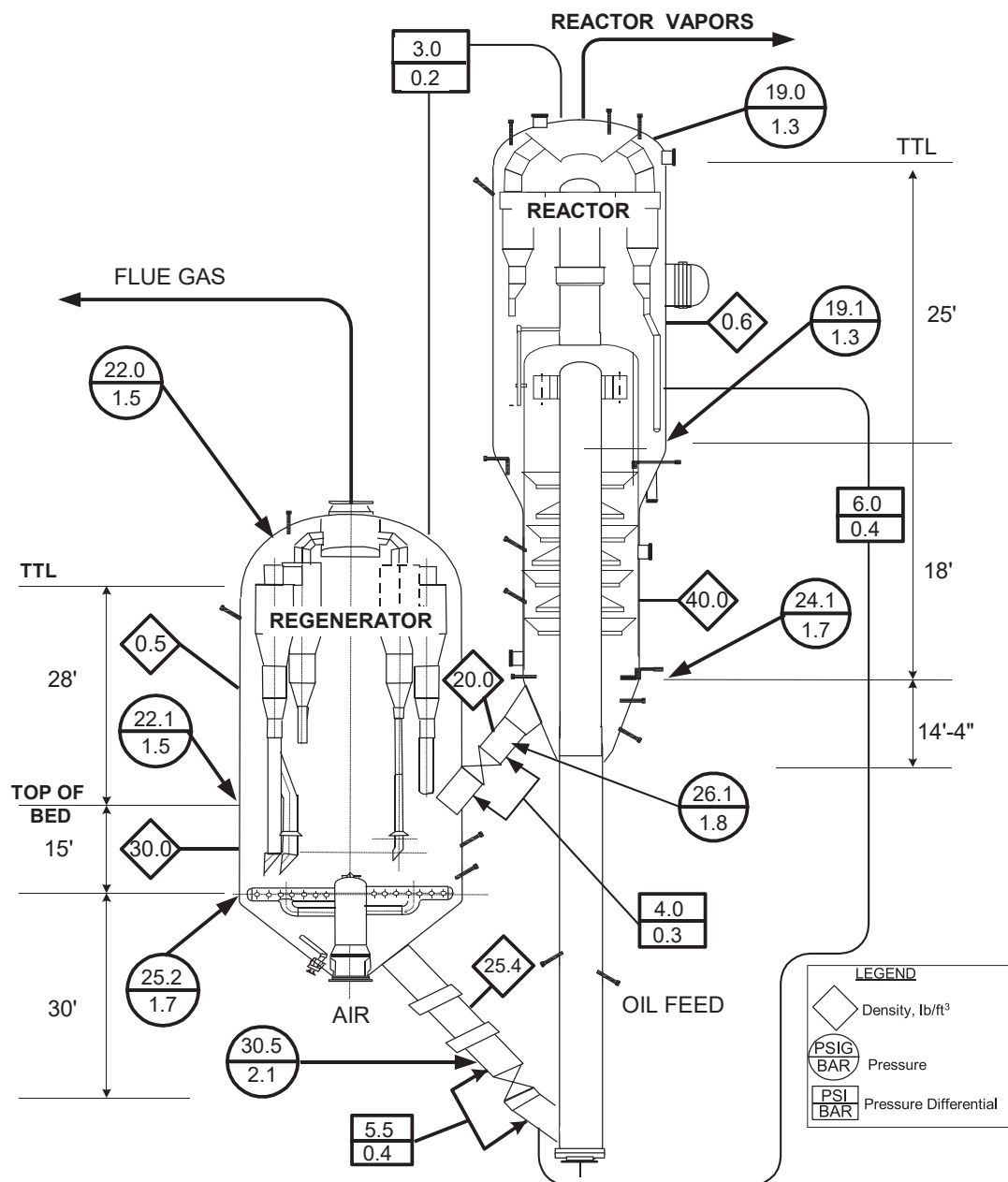
The “ease” of catalyst circulation is largely influenced by the physical layout of the unit and fluidization properties of the catalyst. Some cat crackers circulate with ease regardless of the catalyst’s physical properties. However, in other designs, the unit can experience circulation difficulties with minor changes in catalyst properties.

Things to remember with higher catalyst circulation rate:

- Pressure at the outlet of the regenerated catalyst slide valve goes up, mainly due to higher head pressure and greater friction loss across the J-bend or/Wye-piece section, as well as across the riser. This will result in a lower  $\Delta P$  across the regenerated catalyst slide valve (also see [Example 13.1](#)).
- The higher catalyst circulation rate directionally increases catalyst loss rates from the reactor/regenerator cyclones. This is largely from higher catalyst loading to the cyclones and a higher catalyst attrition rate.
- The performance efficiency of the catalyst stripper goes down due to a “faster flow rate” of catalyst through the stripper. This is particularly true, since most operators do not adjust the stripping steam rate with a higher catalyst circulation rate.
- The higher catalyst circulation rate drags more flue gas into the riser, which can tax the FCC vapor recovery section.
- Long-term, the higher catalyst circulation rate adversely impacts the mechanical reliability of the FCC equipment.

### Example 13.1. Leading indicator regenerated catalyst slide valve opening increase

| Indicator: regenerated catalyst slide valve opening has gradually increased from 40% to 60%                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Evidences                                                                                                                                                                                                                                                     | Possible causes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Field verification: <ul style="list-style-type: none"> <li>• Slide valve <math>\Delta P</math></li> <li>• Reactor-regenerator <math>\Delta P</math></li> <li>• Riser temperature</li> <li>• Regenerator dense bed temperature</li> <li>• Feed rate</li> </ul> | <ol style="list-style-type: none"> <li>1. Pressure above the slide valve is less than typical:               <ul style="list-style-type: none"> <li>• Catalyst is not building enough pressure in the standpipe.</li> <li>• Catalyst entering the standpipe is not properly fluidized.</li> <li>• Not enough aeration along standpipe.</li> <li>• A foreign object restricting catalyst flow at the entrance to standpipe.</li> <li>• A foreign object has fallen into the slide valve.</li> </ul> </li> <li>2. Pressure below the slide valve is higher than normal:               <ul style="list-style-type: none"> <li>• Catalyst is not fully fluidized in the Wye piece or J-bend section.</li> <li>• Coke build-up around the feed nozzles.</li> <li>• Coke in the reactor cyclones.</li> <li>• Coke build-up in the reactor vapor line.</li> <li>• Fouling of main column and/or overhead condensers.</li> </ul> </li> </ol> |

**FIG. 13.1**

Typical FCC unit pressure balance.

Despite the drawbacks noted above, the higher catalyst circulation rate and subsequent higher cat/oil ratio often deliver more liquid volume products from a given FCC feedstock and this often increases the profitability of FCCU operations.

Steady and smooth catalyst circulation increases confidence, as well as the “comfort zone” of the console operator, to optimize the performance of a cat cracker. For example, he or she will be able to:

- Increase feed rate to the unit.
- Increase the stripping steam to reduce carry-under of soft coke and lower the regenerator temperature.
- Reduce the feed preheat temperature to increase cat/oil ratio.
- Increase the cracking temperature to produce more olefin feed and/or increase the gasoline octane.
- Generate more steam from the catalyst cooler.
- Operate at higher CO in the flue gas when operating in partial burn.

Consequently, having the flexibility to maximize the catalyst circulation rate is extremely critical in the long-term reliability and profitability of a given FCC unit.

### 13.3.1 Factors hindering catalyst circulation

The key factors affecting the ability of FCC catalyst to flow smoothly in standpipes are as follows (also see [Example 13.2](#)):

- Condition of the catalyst before it enters the mouth of the standpipe. If the catalyst is not fluidized “correctly”, it is difficult to keep it properly fluidized in the standpipe.
- Depending on the length/height of standpipe, supplemental fluidization must be used to compensate for compression of gas bubbles as it moves downward with the catalyst. The dryness and the amount of the supplemental aeration gas, as well as spacing between the aeration taps are extremely important. In addition, the reliability of measuring the aeration flow rate to each tap, or sets of taps, play key roles in the success of standpipe fluidization. Too much aeration can cause “bridging” of the catalyst flow, and not enough aeration can cause a “stick/slip flow” behavior of the catalyst.
- Catalyst particle size distribution has a huge impact on the ease of catalyst circulation, especially in long standpipes and/or u-bends.

An average standpipe must produce one (1) psig of head pressure per 4 feet of standpipe height (0.07 bar/1.2 m). This pressure gain should be uniform across the entire height of the standpipe. This gain in pressure corresponds to about 35 lb/ft<sup>3</sup> of catalyst flowing density (561 kg/m<sup>3</sup>). There are standpipes in which the catalyst flowing density is in the 45 lb/ft<sup>3</sup> (721 kg/m<sup>3</sup>) range.

#### Example 13.2. Leading indicator erratic catalyst circulation

| Indicator: erratic catalyst circulation                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                   |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Evidences                                                                                                                                                                                                                      | Possible causes                                                                                                                                                                                                                                                                                                                   |
| <ul style="list-style-type: none"> <li>• Low <math>\Delta P</math> alarm coming in and out of fresh feed riser</li> <li>• Fresh feed riser not holding set-point</li> <li>• Regenerator bed temperature is swinging</li> </ul> | <ul style="list-style-type: none"> <li>• Reactor-regenerator pressure balances are off.</li> <li>• Standpipe aeration is not right.</li> <li>• Air rate to the catalyst withdrawal well air rings is not adjusted.</li> <li>• Carbon on cat low or particle size is wrong.</li> <li>• Feed rate or gravity not stable.</li> </ul> |

### 13.4 Catalyst losses

Catalyst losses will have adverse effects on the unit operation, the environment, and operating costs. Catalyst losses appear as excessive catalyst carryover to the main fractionator or losses from the regenerator.

To troubleshoot excessive catalyst losses, one must identify whether the loss is from the reactor (see [Example 13.3](#)) or the regenerator (see [Examples 13.4–13.6](#)). In either case, the following general guidelines should be helpful in troubleshooting catalyst losses:

- Verify the catalyst bed levels in the stripper and regenerator vessels.
- Conduct a single-gauge pressure survey of the reactor-regenerator circuit. Using the results, determine the catalyst density profile and verify the back pressures to the various steam distributors are normal.
- Perform temperature scans to see areas of catalyst de-fluidization.
- Plot the physical properties of the equilibrium catalyst. The plotted properties will include particle size distribution and apparent bulk density. The graph confirms any changes in catalyst properties.
- Have the lab analyze the “lost” catalyst for particle size distribution. The analysis will provide clues as to the sources and causes of the losses.
- Compare the cyclone loading with the design. If the vapor velocity into the reactor cyclones is low, consider adding supplemental steam to the riser. If the mass flow rate is high, consider increasing the feed preheat temperature to reduce catalyst circulation.
- Confirm that the restriction orifices used for instrument purges are in proper working condition and that the restriction orifices are not missing.
- Consider switching to a harder catalyst. For a short-term solution, if the losses are from the reactor side, consider recycling slurry to the riser. If the catalyst losses are from the regenerator, consider recycling catalyst fines to the unit.
- Be prepared to “pressure bump” the reactor or regenerator vessel

#### Example 13.3. Leading indicator catalyst loss from reactor

| Indicator: high ash content—clarified oil (CLO)                                  |                                                                                                                                                                                                                         |
|----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Evidences                                                                        | Possible causes                                                                                                                                                                                                         |
| <ul style="list-style-type: none"><li>• Sample shows high ash content.</li></ul> | <ul style="list-style-type: none"><li>• Reactor level high.</li><li>• Plugged dipleg trickle valve.</li><li>• Hole in dragon head riser termination device (RTD).</li><li>• Blast steam to riser was left on.</li></ul> |

**Example 13.4. Leading indicator catalyst loss from regenerator**

| Indicator: loss of catalyst from regenerator                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Evidences                                                                                                                                                                                                                                                                                                                                                               | Possible causes                                                                                                                                                                                                                                                                                                                                                                                                          |
| <ul style="list-style-type: none"> <li>• Not gaining level in regenerator,</li> <li>• #1 bin at ESP was filling at a faster rate.</li> <li>• Increase in opacity.</li> <li>• Decrease in 0–40 <math>\mu\text{m}</math> fraction of E-cat.</li> <li>• Increase in 80 + micron fraction of E-cat.</li> <li>• Increase in catalyst average particle size (APS).</li> </ul> | <ul style="list-style-type: none"> <li>• Holes in the cyclones.</li> <li>• Holes in the cyclone plenum.</li> <li>• Trickle valve flappers have fallen off.</li> <li>• Catalyst underneath trickle valves is not fluidized.</li> <li>• Catalyst is de-fluidized in the diplegs.</li> <li>• Dipleg diameter is too large.</li> <li>• Refractory lining, hex-steel has fallen off and restricting catalyst flow.</li> </ul> |

**Example 13.5. Leading indicator catalyst loss from regenerator**

| Leading indicator: high flue gas opacity                      |                                                                                                                                                                                                                    |
|---------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Evidences                                                     | Possible causes                                                                                                                                                                                                    |
| <ul style="list-style-type: none"> <li>• DCS alarm</li> </ul> | <ul style="list-style-type: none"> <li>• ESP shutdown (transformer/Rectifier (T/R) failure).</li> <li>• CO boiler shutdown.</li> <li>• Loss of ammonia injection to ESP.</li> <li>• Instrument failure.</li> </ul> |

**Example 13.6. Leading indicator loss of electrostatic precipitator (ESP)**

| Leading indicator: cold temperature on ESP hopper                                                                                                                                              |                                                                                                                                               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Evidences                                                                                                                                                                                      | Possible causes                                                                                                                               |
| <ul style="list-style-type: none"> <li>• Heater not keeping hopper warm (DCS).</li> <li>• No catalyst dropping out when opening hopper valve.</li> <li>• More arcing on hopper T/R.</li> </ul> | <ul style="list-style-type: none"> <li>• Catalyst bridging in hopper.</li> <li>• Hopper plugged.</li> <li>• Hopper heater failure.</li> </ul> |

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## 13.5 Coking/fouling

Nearly every cat cracker experiences some degree of coking/fouling. Coke can be found on the reactor internal walls, reactor top head section, inside/outside of the reactor cyclones, reactor overhead vapor line, main fractionator bottom, and fouling of the slurry bottoms pumparound circuit. Coking and fouling always occur, but they become a problem when they impact throughput or cracking severity.

### 13.5.1 Troubleshooting steps

The following are some of the steps that can be taken to minimize coking/fouling (also see [Examples 13.7 and 13.8](#)):

- *Avoid dead spots.* Coke grows wherever there is a cold spot in the system. Use “dry” dome steam to purge hydrocarbons from the stagnant area above the cyclones. Dead spots cause thermal cracking.
- *Minimize heat losses from the reactor plenum and the transfer line.* Heat loss will cause condensation of heavy components of the reaction products. Insulate as much of the system as possible; when insulating flanges, verify that the studs are adequate for the higher temperature.
- *Improve the feed/catalyst mixing system and maintain a high conversion.* A properly designed feed/catalyst injection system, combined with operating at a high conversion, will crack out high-boiling feeds that otherwise could be the precursors for the formation of coke.
- Ensure cracking temperature is high enough to vaporize/crack very high boiling fraction of the feedstock.
- *Follow proper start-up procedures.* Introduce feed to the riser only when the reactor system is adequately heated up. Local cold spots cause coke to build up in the reactor cyclones, the plenum chamber, or the vapor line.
- *Keep the tube velocity in the bottoms pumparound exchanger(s) greater than 5 ft/s (1.5 m/sec).* Putting the bundles in parallel for more heat recovery may lead to low velocity.
- *Hold the main column liquid bottoms temperature under 700 °F (371 °C).* For residue operation, this temperature should be <650 °F (343 °C). Use “pool quench” to control the main column bottoms temperature.
- Minimize the bottoms level and residence time of the hot liquid.
- Ensure adequate liquid wash to shed trays or grid packing to minimize coking in the bottom of the main column.
- *Utilize a continuous-cycle oil flush into the inlet of the bottoms exchanger.* This keeps the asphaltenes in solution and increases tube velocity.
- *Verify that no fresh feed is entering the main column.* Feed can enter the main column through emergency bypasses, through the feed surge tank vent line or safety relief valve.

**Example 13.7. Leading indicator coking and fouling**

| Leading indicator: high reactor pressure alarm                                                                                     |                                                                                                                                                                                                                                                                                       |
|------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Evidences                                                                                                                          | Possible causes                                                                                                                                                                                                                                                                       |
| <ul style="list-style-type: none"> <li>• DCS trend.</li> <li>• Feed rate had to be cut back.</li> <li>• Blower surging.</li> </ul> | <ul style="list-style-type: none"> <li>• Loss of main fractionator overhead cooling.</li> <li>• Salting of the main fractionator trays or packing.</li> <li>• Coke deposition in the reactor vapor line.</li> <li>• Coke deposition in the rough cut cyclone outlet tubes.</li> </ul> |

**Example 13.8. Leading indicator: coking and fouling**

| Leading indicator: main fractionator slurry pumps cavitations                                                                                                                                                    |                                                                                                                                                                                                                                                   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Evidences                                                                                                                                                                                                        | Possible causes                                                                                                                                                                                                                                   |
| <ul style="list-style-type: none"> <li>• Low slurry pumparound rate.</li> <li>• High slurry pumparound return temperature.</li> <li>• High bottoms temperature.</li> <li>• Reactor pressure climbing.</li> </ul> | <ul style="list-style-type: none"> <li>• Coke buildup in the main fractionator bottom.</li> <li>• Catalyst carryover from the reactor.</li> <li>• Bottoms temperature too high.</li> <li>• Light components entrained with slurry oil.</li> </ul> |

**13.6 Increase in afterburn**

The composition of coke on the spent catalyst is approximately 93% carbon, 7% hydrogen with traces of sulfur and organic nitrogen compounds. It is important that combustion of the coke (see [Table 13.1](#)) occurs in the dense bed of catalyst. Without the catalyst bed to absorb this heat of combustion, the dilute phase and flue gas temperatures increase rapidly, largely from combustion of CO to CO<sub>2</sub>. This phenomenon is called *afterburning*. It is critical that spent catalyst and combustion/lift air are being introduced into the regenerator as evenly as possible across the catalyst bed. It is also important to note that vertical mixing is much faster than lateral mixing.

The magnitude of afterburning in the regenerator largely depends on the operating conditions of the unit and the effectiveness of the contact between the combustion air and the spent catalyst. The geometry of the regenerator and the distribution of the spent catalyst also impact the level of afterburning (see [Example 13.9](#)). Generally speaking, regenerators operating in partial combustion do not experience the same level of afterburning as compared with full-burn regenerators, due to the absence of oxygen in the dilute phase.

Operating options to reduce afterburning include:

- Maximizing the feed preheat temperature.
- Using HCO or slurry oil recycle.
- Optimizing the use of CO combustion promoter.
- Ensuring catalyst circulation from reactor is steady.

- Ensuring catalyst stripping steam rate has been optimized.
- Adjusting combustion air rates to each air distributor.
- Changing the ratio of combustion air and carrier rates.
- Increasing regenerator pressure.
- Increasing the regenerator bed level, while ensuring it does not affect the catalyst loss rate.
- Optimizing the flue gas excess oxygen.

**Table 13.1 Heat of combustion.**

|                                     |   |                  | K cal/kg of C, H <sub>2</sub> , or S | BTU/lb of C, H <sub>2</sub> , or S |
|-------------------------------------|---|------------------|--------------------------------------|------------------------------------|
| C + 1/2 O <sub>2</sub>              | → | CO               | 2200                                 | 3968                               |
| CO + 1/2 O <sub>2</sub>             | → | CO <sub>2</sub>  | 5600                                 | 10,100                             |
| C + O <sub>2</sub>                  | → | CO <sub>2</sub>  | 7820                                 | 14,100                             |
| H <sub>2</sub> + 1/2 O <sub>2</sub> | → | H <sub>2</sub> O | 28,900                               | 52,125                             |
| S + xO                              | → | SO <sub>x</sub>  | 2209                                 | 3983                               |

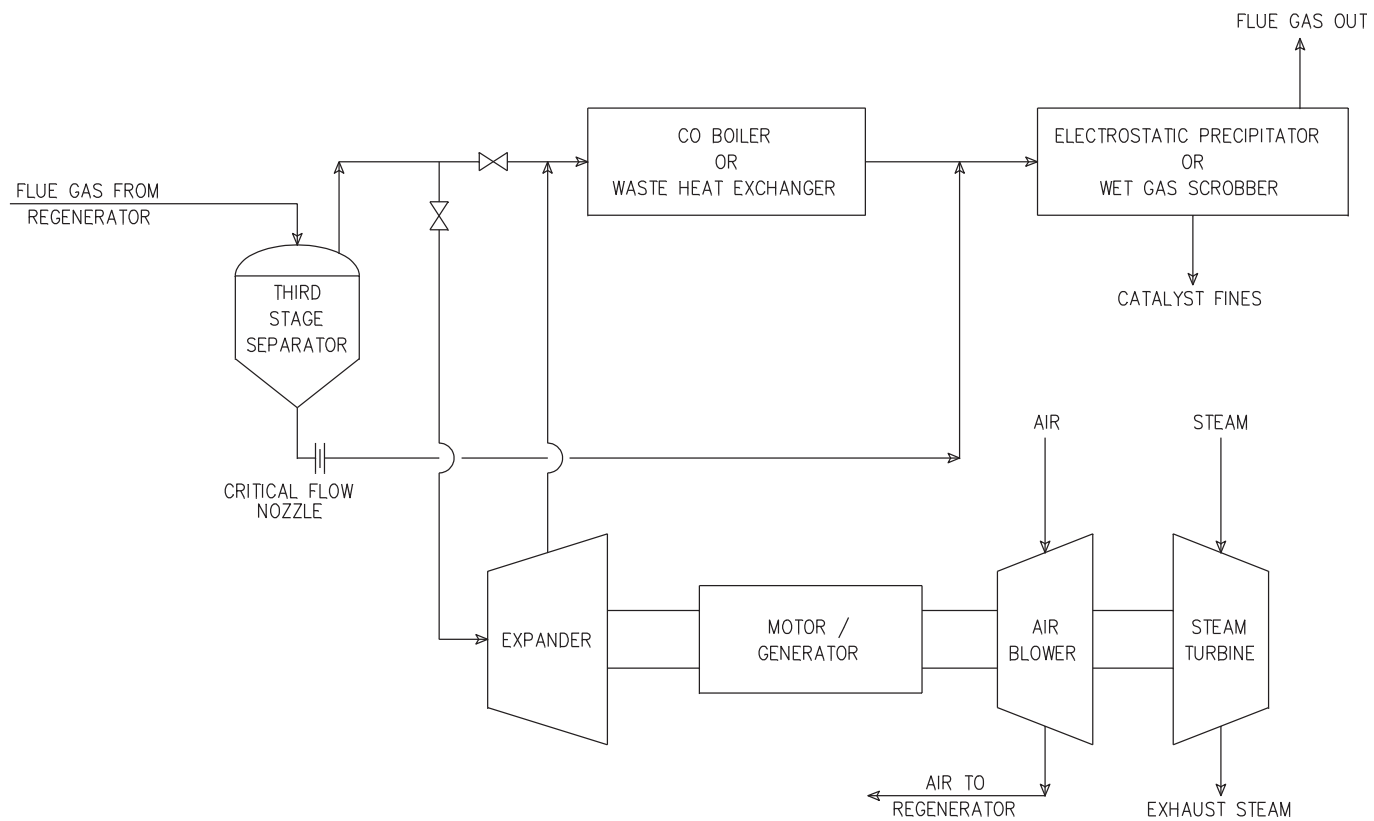
### Example 13.9. Leading indicator: increase in afterburn—high regenerator cyclone outlet temperatures

| Leading indicator: two out of six cyclone outlet temperatures are 50 °F (27.8 °C) higher  |                                                                                                                                                                                                                                                                            |
|-------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Evidences                                                                                 | Possible causes                                                                                                                                                                                                                                                            |
| <ul style="list-style-type: none"> <li>• DCS</li> <li>• feed rate is cut back.</li> </ul> | <ul style="list-style-type: none"> <li>• Broken air distributor arms.</li> <li>• Broken spent catalyst deflector/distributor.</li> <li>• Erratic catalyst flow from stripper.</li> <li>• Low regenerator bed level.</li> <li>• Low regenerator bed temperature.</li> </ul> |

## 13.7 Hot gas expanders

Power recovery trains recover energy from the flue gas (see Fig. 13.2 for a typical flue gas power recovery scheme). The FCC starts to resemble a large jet engine; air is compressed into a combustion zone and expanded across a turbine. Power recovery increases the efficiency of the unit but adds one more mechanical device to an already long list. Since they are too big to bypass, power trains need to be as reliable as the rest of the unit.

The main concerns in the design and operation of a power recovery system are catalyst fines and temperature. Catalyst fines will lead to serious blade wear, deposits, power loss, and rotor vibration. Deposit occurs mostly where flue gas velocities are at maximum levels, such as the blade outer diameter (also see Examples 13.10 and 13.11).



**FIG. 13.2**

Typical flue gas recovery scheme.

**Example 13.10. Leading indicator: hot gas expander**

| Leading indicator: loss of horsepower                                                                                                                                                             |                                                                                                                                                         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| Evidences                                                                                                                                                                                         | Possible causes                                                                                                                                         |
| <ul style="list-style-type: none"> <li>• More supplemental steam is needed.</li> <li>• Suction butterfly valve is “more closed.”</li> <li>• Expander outlet temperature has increased.</li> </ul> | <ul style="list-style-type: none"> <li>• Rotor blade erosion.</li> <li>• Critical flow nozzle damaged.</li> <li>• Bypass valve is 100% open.</li> </ul> |

**Example 13.11. Leading indicator: hot gas expander**

| Leading indicator: expander vibration has increased                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Evidences                                                                                                                                                                                                                     | Possible causes                                                                                                                                                                                                                                                                                                                                                                     |
| <ul style="list-style-type: none"> <li>• DCS trend.</li> <li>• Field verification.</li> <li>• Higher stack opacity.</li> <li>• Higher catalyst loss from regenerator.</li> <li>• Increase in fresh catalyst usage.</li> </ul> | <ul style="list-style-type: none"> <li>• Catalyst built up on the shroud.</li> <li>• Disc failure from intergranular sulfidation attack.</li> <li>• 3rd-stage separator not working properly.</li> <li>• Soft catalyst.</li> <li>• High concentration of sodium, vanadium, magnesium, iron, or calcium on the catalyst.</li> <li>• Catalyst being attritted prematurely.</li> </ul> |

**13.7.1 Troubleshooting steps**

1. Regular monitoring of rotor blade conditions by visual inspection, photographs, and/or video recording. A port is usually installed for this.
2. Continuous monitoring of rotor casing vibration, bearing temperatures, and the expander inlet/outlet temperatures. Problems can be either instantaneous or slow-growing. Instantaneous problems occur during startup, upset, and shutdown, and are easy to note. Slow-growing problems can creep up and are almost invisible, while everything is running well. Compare the readings month-to-month to spot trends.
3. Continuous monitoring of the third-stage separator performance. If catalyst is showing up downstream, consider using more than the “standard” 3% flue gas underflow. The blowcase needs more attention than it usually gets.
4. On-line cleaning—injecting of walnut hulls into the inlet of the expander weekly.

5. Thermal shocking—reduce feed in 20% increments, while maintaining maximum air rate to the regenerator. Cool the expander inlet temperature to around 1000 °F (540 °C) and hold for at least 1 h. This is not a procedure that the expander vendor supports, but it is practiced by many refiners.

## 13.8 Flow reversal

A stable pressure differential must be maintained across the slide valves. The direction of catalyst flow must always be from the regenerator to the reactor and from the reactor-stripper back to the regenerator. A negative differential pressure across the regenerated catalyst slide valve can allow fresh feed and oil-soaked catalyst to backflow from the riser into the regenerator. This flow reversal can result in uncontrolled burning in the regenerator and potentially damage the regenerator internals due to the extremely high temperature, costing a refiner several million dollars in production loss and maintenance expense.

Similarly, a negative pressure differential across the spent catalyst slide valve can allow hot flue gas to backflow to the reactor and the main fractionator, severely damaging the mechanical integrity of these vessels.

Some of the main causes of loss of pressure differential across the slide valves are as follows:

- Loss of the main air blower (MAB) or the wet gas compressor (WGC).
- Loss of the catalyst cooler.
- Presence of water in the feed.
- High catalyst circulation rates, resulting in excessive slide valve opening and low differential.
- Loss of regenerator or reactor stripper bed levels.
- Failure of the reactor temperature controller and reactor-stripper level controller.
- Bypass open around a shutdown valve.

### 13.8.1 Reversal prevention philosophy

The FCC process is very complex and many scenarios can upset operations. If the upset condition is not corrected or controlled, each scenario could lead to a flow reversal. [Table 13.2](#) contains a cause/effect shutdown matrix indicating scenarios in which a shutdown (reversal) could take place. In most cases, a unit shutdown is not necessary if adequate warning (low alarms before low/low shutdowns) is provided. The operating staff must be trained to respond to those warnings.

The shutdown system will have adequate interlocks to prevent inadvertent trips. The system must include “two-out-of-three voting” (2oo3) or backup instruments. The operators must trust the system for it to stay in service.

Slide valves will have an independent low differential pressure override controller to prevent the reactor temperature controller from opening the slide valves to the point where low differential pressure could allow feed back to the regenerator.

**Table 13.2 A cause and effect shutdown matrix.**

| Cause<br>↓                                            | Effect → | Close riser<br>regenerated<br>catalyst<br>slide valve | Open riser<br>emergency<br>steam<br>valve | Close<br>feed to<br>riser | Close<br>slurry<br>recycle<br>valve | Close<br>HCO<br>recycle<br>valve | Close<br>spent<br>catalyst<br>slide<br>valve | Open<br>regenerator<br>emergency<br>steam valve | Alarm<br>only |
|-------------------------------------------------------|----------|-------------------------------------------------------|-------------------------------------------|---------------------------|-------------------------------------|----------------------------------|----------------------------------------------|-------------------------------------------------|---------------|
| Regenerated<br>catalyst slide valve<br>low $\Delta P$ |          |                                                       |                                           |                           |                                     |                                  |                                              |                                                 | X             |
| Spent catalyst slide<br>valve low $\Delta P$          |          |                                                       |                                           |                           |                                     |                                  |                                              |                                                 | X             |
| Air blower low/low<br>air flow                        |          | X                                                     | X                                         | X                         | X                                   | X                                | X                                            | X                                               |               |
| Riser low/low feed<br>flow rate                       |          | X                                                     | X                                         | X                         | X                                   | X                                | X                                            | X                                               |               |
| Low reactor<br>temperature                            |          |                                                       |                                           |                           |                                     |                                  |                                              |                                                 | X             |
| Reactor vessel high<br>catalyst level                 |          |                                                       |                                           |                           |                                     |                                  |                                              |                                                 | X             |
| Manual shutdown                                       |          | X                                                     | X                                         | X                         | X                                   | X                                | X                                            | X                                               |               |

### Example 13.12. Common scenarios/leading indicators.

|   | Leading indicators                                               | Confirming evidence                                                                                                                                                                                                                                                       | Probable causes                                                                                                                                                                                                                              |
|---|------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Loss of charge (instrument freeze up).                           | <ul style="list-style-type: none"> <li>• Increase in heater temperature.</li> <li>• Regenerator bed temperature increase.</li> <li>• Reactor top temperature increase.</li> <li>• Too much compressor suction pressure.</li> </ul>                                        | <ul style="list-style-type: none"> <li>• Loss of charge pump.</li> <li>• Water in charge.</li> <li>• Instrument failure (freeze up).</li> </ul>                                                                                              |
| 2 | Feed preheater control valve went open to 100%.                  | <ul style="list-style-type: none"> <li>• DCS feedback</li> <li>• Verification of control valve position.</li> <li>• Feed pump operation.</li> <li>• Feed preheater pressure increase.</li> </ul>                                                                          | <ul style="list-style-type: none"> <li>• Water in feed.</li> </ul>                                                                                                                                                                           |
| 3 | Loss of main air blower.                                         | <ul style="list-style-type: none"> <li>• Unit gets quiet.</li> <li>• Riser slumps.</li> </ul>                                                                                                                                                                             | <ul style="list-style-type: none"> <li>• Steam issues - wet steam.</li> <li>• Lube oil system for main air blower.</li> <li>• Loss of oil pressure.</li> <li>• Vacuum issues.</li> </ul>                                                     |
| 4 | Started losing vacuum on blower.                                 | <ul style="list-style-type: none"> <li>• Loss of vacuum (DCS).</li> </ul>                                                                                                                                                                                                 | <ul style="list-style-type: none"> <li>• Heavy rain.</li> <li>• Steam pressure drop.</li> <li>• Loss of sealing steam.</li> <li>• Vacuum leak on pump.</li> <li>• Hot well level.</li> <li>• Surface condenser pump failure.</li> </ul>      |
| 5 | Riser temperature falls below 00 °F (482 °C).                    | <ul style="list-style-type: none"> <li>• Low temperature DCS alarm.</li> <li>• Regenerator slide valve wants to open.</li> <li>• Delta P dropping on regenerator slide valve.</li> <li>• No automatic feed diversion.</li> </ul>                                          | <ul style="list-style-type: none"> <li>• Bad riser temperature indicator (TI).</li> <li>• Reduced catalyst circulation (or none).</li> <li>• No catalyst cracking.</li> <li>• Feed oil going in with no catalyst.</li> </ul>                 |
| 6 | Regenerator temperature increase of 6 °F (3.3 °C).               | <ul style="list-style-type: none"> <li>• Compare other regenerator TI's.</li> <li>• Regenerator overhead temperature is up.</li> <li>• Slide valve position has decreased.</li> <li>• Catalyst circulation has decreased.</li> <li>• Conversion has decreased.</li> </ul> | <ul style="list-style-type: none"> <li>• Poorer feedstock.</li> <li>• O<sub>2</sub> controller backed off from partial to full combustion.</li> <li>• High feed preheat temperature.</li> <li>• Fresh catalyst addition too high.</li> </ul> |
| 7 | Regenerator dilute phase temperature increase of 40 °F (22.3 °C) | <ul style="list-style-type: none"> <li>• Regenerator temperature profile.</li> <li>• Main air blower speed.</li> <li>• Air ring position.</li> <li>• Flue gas analyzers (O<sub>2</sub> and CO).</li> </ul>                                                                | <ul style="list-style-type: none"> <li>• Air/catalyst mal-distribution.</li> <li>• High air rate.</li> <li>• Possible broken air ring.</li> <li>• Transition between burns.</li> <li>• Mechanical failure.</li> </ul>                        |

*Continued*

| —cont'd |                                                               |                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                |
|---------|---------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | Leading indicators                                            | Confirming evidence                                                                                                                                                                                                                                                                  | Probable causes                                                                                                                                                                                                                                                                |
| 8       | Carbon on catalyst is going up from 0.1 to 0.3 wt%.           | <ul style="list-style-type: none"> <li>• Regenerator operating temperature.</li> <li>• Lab data—visual inspection of catalyst.</li> <li>• Reactor stripper operations are poor.</li> <li>• Regenerator flue gas analyzer has changed.</li> </ul>                                     | <ul style="list-style-type: none"> <li>• Change in feed quality.</li> <li>• Main air blower flow has decreased.</li> <li>• Lack of enriched oxygen feed has decreased.</li> <li>• Poor hydrocarbon stripping in reactor stripper.</li> <li>• Too much Sox additive.</li> </ul> |
| 9       | Flue gas CO exceeded permit limit of 500 ppm for 3 h          | <ul style="list-style-type: none"> <li>• CO analyzer</li> <li>• Lower regenerator dense bed temperature by 18 °F (10 °C)</li> </ul>                                                                                                                                                  | <ul style="list-style-type: none"> <li>• Slurry recycle to riser got plugged</li> <li>• Too much slurry recycle to the riser</li> <li>• Loss of feed preheat furnace</li> </ul>                                                                                                |
| 10      | Loss slurry pump during unit start-up                         | <ul style="list-style-type: none"> <li>• Could not get the slurry out of the tower</li> <li>• Tank level not rising</li> <li>• Pumps were dead heading</li> </ul>                                                                                                                    | <ul style="list-style-type: none"> <li>• Bad level indication</li> <li>• Catalyst carryover from reactor vessel</li> <li>• Too light of slurry oil</li> </ul>                                                                                                                  |
| 11      | Regenerator slide valve opening increased from 40% to 60%.    | <ul style="list-style-type: none"> <li>• Riser temperature lowered.</li> <li>• Decrease in feed temperature.</li> <li>• Field verify valve opening %.</li> </ul>                                                                                                                     | <ul style="list-style-type: none"> <li>• Feed preheater tripped.</li> <li>• Emergency steam flow control valve (FCV) failed to open.</li> <li>• Catalyst bridged above slide valve in standpipe.</li> </ul>                                                                    |
| 12      | Spent catalyst slide valve opening increased from 30% to 60%. | <ul style="list-style-type: none"> <li>• Reactor bed level.</li> <li>• Reactor pressure taps.</li> <li>• Valve position indicator.</li> <li>• Pressure change.</li> </ul>                                                                                                            | <ul style="list-style-type: none"> <li>• Reactor level high due to bridging.</li> <li>• Plugged taps in regenerator.</li> <li>• Trash in unit.</li> </ul>                                                                                                                      |
| 13      | Flue gas slide valve position increased from 35% to 65%.      | <ul style="list-style-type: none"> <li>• Delta P across slide valve has dropped.</li> <li>• Field-verify position of valve.</li> <li>• Regenerator pressure increase.</li> </ul>                                                                                                     | <ul style="list-style-type: none"> <li>• Catalyst restrictions.</li> <li>• Orifice chamber malfunction.</li> </ul>                                                                                                                                                             |
| 14      | High CO boiler firebox temperature.                           | <ul style="list-style-type: none"> <li>• Low dense bed temperature.</li> <li>• High CO carryover.</li> <li>• High supplemental fuel gas flow.</li> </ul>                                                                                                                             | <ul style="list-style-type: none"> <li>• Lost air to regenerator (compressor).</li> <li>• Lost instrumentation on fuel gas.</li> <li>• Poor unit conversion.</li> </ul>                                                                                                        |
| 15      | CO boiler firebox temperature increase of 30 °F (16.7 °C).    | <ul style="list-style-type: none"> <li>• Increase in steam production.</li> <li>• Increase in flue gas temperature from CO boiler outlet.</li> <li>• Increase in CO level in regenerator.</li> <li>• Flue gas slide valve has increased opening.</li> <li>• Feed quality.</li> </ul> | <ul style="list-style-type: none"> <li>• Lower API feed quality.</li> <li>• Lower oxygen from main air blower.</li> <li>• Not enough stripping steam.</li> <li>• Catalyst addition—issues, losses.</li> <li>• Fuel gas quality.</li> <li>• Drop in feed temperature</li> </ul> |
| 16      | Poor unit conversion.                                         | <ul style="list-style-type: none"> <li>• Lab data.</li> <li>• Change in hydrocarbon cuts on fractionator side.</li> <li>• Change in carbon on regenerated catalyst (CRC).</li> </ul>                                                                                                 | <ul style="list-style-type: none"> <li>• Too low combustion air rates.</li> <li>• Charge make up changed.</li> <li>• Improper reflux rates.</li> </ul>                                                                                                                         |

|    |                                                          |                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                              |
|----|----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 17 | Rise in clarified oil gravity from -4 to 0 °API          | <ul style="list-style-type: none"> <li>• Lab analysis.</li> <li>• CLO flow increase.</li> <li>• LCO and HCO flow decrease.</li> <li>• Main fractionator level increase.</li> </ul>                                                                                                                            | <ul style="list-style-type: none"> <li>• Main fractionator temperature control problem.</li> <li>• Feed temperature, conversion, catalyst activity.</li> <li>• Feed quality decrease.</li> <li>• Slurry exchangers have uneven flow.</li> </ul>                                                                              |
| 18 | Clarified oil (CLO) gravity went to 0 or positive °API.  | <ul style="list-style-type: none"> <li>• Lab results.</li> <li>• Main fractionator level rising.</li> <li>• Temperature on main fractionator.</li> </ul>                                                                                                                                                      | <ul style="list-style-type: none"> <li>• CLO make is too great.</li> <li>• Uncracked feed to main fractionator.</li> <li>• Feed composition.</li> <li>• Main fractionator bottom temperature too low.</li> <li>• Riser temperature low.</li> </ul>                                                                           |
| 19 | High decant oil make rate.                               | <ul style="list-style-type: none"> <li>• Decant oil flow rate is high.</li> <li>• APC system misses target rate.</li> <li>• Mass balance indications.</li> <li>• Main fractionator bottoms level high/low temperature.</li> <li>• Decant oil API gravity high.</li> <li>• Pumparound flow is high.</li> </ul> | <ul style="list-style-type: none"> <li>• Riser temperature is low.</li> <li>• Feed preheat temperature is high.</li> <li>• Main fractionator pressure is high.</li> <li>• Feed composition has changed (worse).</li> <li>• Catalyst activity is down.</li> <li>• Feed drum is relieving to the main fractionator.</li> </ul> |
| 20 | Clarified oil (CLO) gravity increase from -4 to +2 °API. | <ul style="list-style-type: none"> <li>• CLO rate increase.</li> <li>• Fractionator bottoms level increase.</li> <li>• HGO cutter rate increase.</li> </ul>                                                                                                                                                   | <ul style="list-style-type: none"> <li>• Main fractionator bottoms X-ray level control malfunction.</li> <li>• LCO purge to pump.</li> <li>• Feed to main fractionator fails.</li> </ul>                                                                                                                                     |
| 21 | Main fractionator loss of level.                         | <ul style="list-style-type: none"> <li>• Level indicators (DCS and outside).</li> <li>• No pumparounds.</li> <li>• Main fractionator high temperature.</li> <li>• Main fractionator delta P lower</li> </ul>                                                                                                  | <ul style="list-style-type: none"> <li>• Pump out is open.</li> <li>• Started with no level.</li> <li>• Hot reactor overhead to main fractionator with no pumparounds.</li> <li>• Operator experience level low.</li> </ul>                                                                                                  |
| 22 | Loss of flow through slurry exchangers.                  | <ul style="list-style-type: none"> <li>• Decrease flow rates (U factors, DCS).</li> <li>• Low flow to debutanizer reboiler.</li> </ul>                                                                                                                                                                        | <ul style="list-style-type: none"> <li>• Catalyst carryover.</li> <li>• Polymer buildup.</li> </ul>                                                                                                                                                                                                                          |
| 23 | Dry gas yield increase of 20%                            | <ul style="list-style-type: none"> <li>• Compressor amp load increases (DCS).</li> <li>• Reactor temperature increases.</li> <li>• Regenerator pressure increases.</li> <li>• Dry gas scrubber delta P increases.</li> </ul>                                                                                  | <ul style="list-style-type: none"> <li>• Feed change (lower API Gravity) aniline point decrease, higher metals content.</li> <li>• Regenerator dense bed temperature increased.</li> <li>• Catalyst flow rate increased.</li> </ul>                                                                                          |
| 24 | Dry gas yield increase from 10% to 15%                   | <ul style="list-style-type: none"> <li>• Lab analysis—check C<sub>3</sub>'s.</li> <li>• Absorber flow.</li> <li>• Wet gas compressor.</li> <li>• Absorber temperature profile.</li> </ul>                                                                                                                     | <ul style="list-style-type: none"> <li>• Increased reactor outlet temperature.</li> <li>• High regenerator temperature.</li> <li>• Amount of dispersion steam.</li> <li>• Riser outlet temperature (ROT).</li> </ul>                                                                                                         |

Continued

| —cont'd |                                                              |                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------|--------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | Leading indicators                                           | Confirming evidence                                                                                                                                                                                                                                                                                                                 | Probable causes                                                                                                                                                                                                                                                                                                                                                                                                       |
| 25      | Wet gas compressor (WGC) is at its capacity limit.           | <ul style="list-style-type: none"> <li>• High first stage deviation.</li> <li>• High off-gas make rate.</li> <li>• Poor unit conversion.</li> </ul>                                                                                                                                                                                 | <ul style="list-style-type: none"> <li>• High metals concentration on catalyst.</li> <li>• Not enough gasoline condensing in main column overhead receiver.</li> <li>• Hydrogen dumping from gasoline sulfur removal unit to cold receiver.</li> </ul>                                                                                                                                                                |
| 26      | Loss of first stage wet gas compressor.                      | <ul style="list-style-type: none"> <li>• High pressure in first stage KO drum.</li> <li>• Main fractionator pressure up.</li> <li>• Reactor pressure up.</li> <li>• Slide valve delta P low.</li> <li>• Regenerator pressure up.</li> <li>• Second stage compressor trip.</li> </ul>                                                | <ul style="list-style-type: none"> <li>• Lube oil pump trip.</li> <li>• First stage suction KO drum level high.</li> <li>• First stage compressor vibration trip.</li> <li>• Loss of power to Bentley Nevada shutdown system.</li> </ul>                                                                                                                                                                              |
| 27      | Debutanizer bottoms temperature decrease by 30 °F (16.7 °C). | <ul style="list-style-type: none"> <li>• HCO flow/temp to debutanizer reboiler.</li> <li>• HCO to debutanizer reboiler control valve position.</li> <li>• Debutanizer temperature profile.</li> <li>• HCO pumparound system.</li> <li>• Main fractionator temperature profile.</li> <li>• Debutanizer analyzers delta P.</li> </ul> | <ul style="list-style-type: none"> <li>• Loss of HCO pump.</li> <li>• Control valve failure (HCO to debutanizer reboiler).</li> <li>• Main fractionator upset—loss of slurry pump resulting in loss of HCO tray level.</li> <li>• Debutanizer tower mechanical issue.</li> <li>• Rapid feed increase to debutanizer.</li> <li>• Debutanizer reflux rate increase.</li> <li>• Water in feed to debutanizer.</li> </ul> |
| 28      | Debutanizer bottoms temperature falls 60 °F (33.3 °C).       | <ul style="list-style-type: none"> <li>• Tower level went up.</li> <li>• Tower top temp went down.</li> <li>• Level in overhead drum went down.</li> <li>• RVP in gasoline went up.</li> </ul>                                                                                                                                      | <ul style="list-style-type: none"> <li>• Loss of 600# steam.</li> <li>• Loss of HCO.</li> <li>• Control valve problems.</li> <li>• HCO pump problems.</li> </ul>                                                                                                                                                                                                                                                      |
| 29      | High pressure in debutanizer tower during startup.           | <ul style="list-style-type: none"> <li>• Tower bottoms temperature cold on debutanizer and stripper.</li> <li>• Absorber pressure high.</li> <li>• High C<sub>2</sub>'s on analyzer (absorber stripping tower).</li> </ul>                                                                                                          | <ul style="list-style-type: none"> <li>• Absorber stripping tower cold.</li> <li>• No heat medium flow.</li> <li>• Absorber stripping tower pressure high.</li> </ul>                                                                                                                                                                                                                                                 |
| 30      | “Fly wheeling” of H <sub>2</sub> S stripper (de-ethanizer).  | <ul style="list-style-type: none"> <li>• Increase of feed to stripper.</li> <li>• High delta P across the stripper</li> <li>• High liquid level in the high pressure separator.</li> </ul>                                                                                                                                          | <ul style="list-style-type: none"> <li>• High pressure receiver is too cold.</li> <li>• Not enough reboiling.</li> </ul>                                                                                                                                                                                                                                                                                              |

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## Summary

This chapter emphasizes that effective and timely troubleshooting largely depend on being extremely familiar with “normal” conditions as they relate to the feedstock quality, catalyst properties, operating conditions, reactor yields, pressure balance and equipment performance parameters. This chapter also provides examples of common problems, symptoms, and probable causes that one may encounter in troubleshooting FCC units. In addition, a systematic approach is outlined to provide solutions and corrective action. The suggested solutions are necessarily generic but apply to a wide variety of units.

In closing, the remaining pages contain examples of actual events as described by FCC operators during my customized training classes. These case studies can be used as a guide to troubleshoot similar events.